

# ***USER REQUIREMENT SPECIFICATION***

## ***(SECTION: FORMULATION)***

|                       |                                 |
|-----------------------|---------------------------------|
| <b>DEPARTMENT</b>     | <b>FORMULATION DEVELOPEMENT</b> |
| <b>EQUIPMENT NAME</b> | <b>COATER</b>                   |
| <b>DOCUMENT No</b>    | <b>UR/FD/404</b>                |

|                                                         |        |                            |
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**(A) OBJECTIVE/ PREAMBLE:**

- To establish the documented evidence to specify the user specific requirements of Auto-coater (Coating Machine) with precise controls to perform Functional/Drug/Sugar/Pellet Coating.
- Capacity of Proposed Auto Coater 50 gm to 1 kg.
- Equipment shall operate under conditions as per the GMP requirements.
- Auto Coater should have capability to execute Process Optimization/DOE Trials on Scale-Down Version of Production Model, to define Critical Process Parameters (CPP) and develop Design Space in line with QbD Principles.
- Different coating should be possible with proposed equipments like film coating, **sugar coating**, functional coating, **pellets** coating and drug loading on tablets. Sugar Coating Not Required
- For coating, the pan should have variable speed with baffle inside for uniform mixing of tablets. The equipment should facilitate the coating of **moisture sensitive products** without affecting its integrity. Minitabs Not Pellets: Client to confirm size  
OPTION FOR DEHUMIDIFIER DX TYPE AND CHILLED WATER RECIRCULATION TYPE TO BE GIVEN

**(B) SCOPE:**

This requirement of auto coater is for coating area in formulation Development area for taking pilot scale batch ranging from 0.1 kg to 1 kg. The URS is provided to help the supplier to identify the important and variable component with other optional requirement.

**(C) PROCESS/PRODUCT REQUIREMENTS:**

**1.0 Description of the equipment:**

**Tablet Coater:**

- The required equipment is for film coating/ Functional coating/ Drug coating of the tablets in a closed system. The purpose of the tablet coating system is to coat tablets uniformly by spraying them with coating materials in controlled environment & on a moving tablet bed. The hot air entering this equipment will enable a very efficient drying of tablets, while they are being coated. Complete system comprise of sub sections: Pan Housing (Inlet Air, Pan & Spraying System) and Exhaust Air handling System with **Scrubber**. OPTION FOR WET SCRUBBER AND EXHAUST 10MICRON POLICE FILTER TO BE GIVEN

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- All stainless steel construction ,316L product contact
- Sealed access doors , self draining cabinet
- Interchangeable pan sizes (0.100 kg to 1 kg)
- Quick release supply exhaust plenums
- Flame proof Cabinet process light FLAMEPROOF LAMP REQUIRED
- Provision of sodium vapour lamp/Light filters for light sensitive drugs NOT REQUIRED
- Tablet sampling probe –Auto sampling should be possible
- Cabinet drain c/w manual drain valve
- Double wall insulated cabinet on front , rear , side walls and doors NOT REQUIRED

### Coating Pan:

Different size of coating pan to cater different batch size 50 gm , 100gm,200gm,500gm an 1 kg

### Spray Systems:

- Guns Make: Schlick or other standard make spray nozzles, with suitable nozzle diameter for Different pan size with atomizing and pattern air control HUGOPHARM MAKE NOZZLE 0.8mm / 1mm Spare TO BE QUOTED
- Adjustable nozzle position with marking
- Fitting, tubing, atomizing and pattern air regulators
- Peristaltic Pump : Suitable to accommodate spray rate form 10 gm/ml to 200 gm/ml
- Spray Rate: To achieve spray rate from 10-200 gm/ml for coating.
- Electrically operated Peristaltic Dosing Pump having silicon tubing shall be provided for product dosing 1Nos of air borne spray guns having air inlet and liquid connection mounted on specially designed header

### Air Handling Units:

- Hot Air System: Air Inlet shall be having 5-micron filter. Air Blower shall be driven 0.25 HP 230 V electric motor Air Inlet shall be having 5-micron filter. Air Blower shall be driven 0.25 HP 230 V electric motor
- Heating shall be done with electrical heater having heating load of 3 K.W. Temperature range 25°to 85°C at coater
- Exhaust System: Exhausting of air from the pan shall be done with air blower that shall be driven with 0.5 HP 230 V electric motor.

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### Control panel :

The control panel comprise of the following:

1. Hot Air Blower RPM – Indication & Control
2. Exhaust Air Blower RPM - Indication & Control
3. Pan Motor RPM – Indication & Control
4. Inlet Air Temperature – Indication & Control
5. Exhaust Air Temperature- Indication
6. Product Temperature – Indication
7. Atomization Pressure – Indication & Control

FAN AIR CONTROL AND INDICATION TO BE PROVIDED

### 2.0 Purpose of equipment:

- Purpose of the equipment for scale up batch from 50 gm to 1 kg
- Equipments should coat different sized and shaped tablets using aqueous or non-aqueous Coating solution/suspension with or without colours.
- Equipment shall operate under conditions as per the GMP requirements.
- For coating, the pan should have variable speed with baffle inside for uniform mixing of tablets. The equipment should facilitate the coating of moisture sensitive products without affecting its integrity.
- The equipment shall be suitable for film /Functional /**Sugar coating**/drug coating of the tablets.

SUGAR COATING NOT REQUIRED

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### 3.0 Number of equipments with required capacity details :

- Number of equipment - 1 no.
- Product dosage type : Oral solid dosage form

### 4.0 Suggested supplier's name:

Hugopharma

### 5.0 Requirements for any power failure backup's or other utilities:

Not applicable

### 6.0 Any other specific requirement:

- Baffle Type: It shall provide efficient mixing of tablets for film coating ,Functional coating Sugar coating and API coating
- Up gradation provision for API coating and sugar coating SUGAR COATING NOT REQUIRED
- Up gradation provision for pellets coating of size of 500 micron to 2000 micron Miniballs Not Pellets: Client to confirm size
- Modification in exhaust blower should be possible for pellets coating
- Inter connectivity of balance with coater to measure spray rate.
- All the parameters including alarms, filters will be monitored through the control panel.
- Provision for PAT tool application. NOT REQUIRED
- Flame proofing features
- Flame proof pump motor NOT REQUIRED
- Explosion resistant dust collector with grounded media filters NOT REQUIRED
- Static ground wire and lugs on system where necessary
- Explosion proof process light for coater
- RTD for the measurement of bed temperature
- Accessories like Pipes , Clamps etc required for connection and fitting with AHU and coating machine

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**(D) OPERATIONAL REQUIREMENTS :****1.0 Production Stage :**

- Auto coater shall uniformly coat tablets/pellets by spraying them with coating materials in controlled environment on a moving tablet bed. The hot air entering this
- Machine shall not contribute towards any edge erosion or breaking of tablets and there shall be minimal loss of coating materials.
- Pan surface shall be smooth with RA 0.4  $\mu\text{m}$
- Should be minimum process loss when drug coating is applied in solution /suspension form on tablets

**2.0 Material inputs into the equipment :**

- Core tablets / pellets of different size and shape. Minitabs Not Pellets: Client to confirm size
- Coating materials used may be of different chemical nature –colours /organic solvent.

**3.0 Desired out come from the equipment :**

- Uniform coated tablets / pellets.
- Shade/colour is controlled and uniform.
- Uniform weight gain during coating process
- Elegant appearance of product.
- Uniformity of content when drug coating done on tablets

**4.0 Charge size / Lot size:**

Maximum up to: 1kg

**5.0 Method of cleaning :**

Any of the following methods can be used: DRY CLEANING TO BE DONE

- By filling the water in the pan and rotating the pan. NO CIP REQUIRED
- By filling the alkaline solution in the pan and rotating the pan.
- By manually scrubbing using nylon scrubber with water.

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**6.0 Recommended solvent for cleaning :**

- Purified water.
- Alkali solution for enteric coated tablets.

**7.0 Method for loading material into the equipment :**

- Manually

**8.0 Method for unloading material from the equipment :**

- It should be manually from the product containers. **DISCHARGE SHALL BE MANUAL**
- Design of coater & baffles should be such that no tablet should remain stagnant between pan and baffles while unloading; even last tablet should get discharged.

**9.0 Expected running time per day :**

- 3 shifts in a day (8 hours / Shift)

**10.0 Utilities requirement :**

Following Utilities shall be provided by TPL

- Steam
- Compressed Air
- Electrical connection as per equipment requirement.
- Prewash water

**11.0 Instrumentations :****(1) List of instruments associated with equipment**

- Shall be provided by vendor.



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**(2) Automation:**

Machine shall be provided with suitable operator interface terminal (OIT) to operate following process parameters.

- Inlet Air Fan On/off
- Exhaust Air Fan On/Off
- Spray Nozzles On/Off
- Inlet air temperature-set & actual with P&I Loop.
- % RH display
- Exhaust air temperature-actual
- Differential Pressure across HEPA with set point.
- Differential Pressure across the Pan
- Air Flow CFM actual & set point INDICATION ONLY
- Dew point display
- Different Safety Alarms interlocks
- Process On/Off
- Setting of spray on/off & timing control
- Setting of Pan RPM & controls

**12.0 Change over parts requirements :**

- As per vendor recommendation

**13.0 Factory Acceptance Test**

- Supplier to arrange FAT at their respective manufacturing site
- Supplier to demonstrate all functions of equipment, instrument and controls

**14.0 Training :**

- Site training with regard to operation, cleaning and maintenance

**15.0 Any other specific requirements.**

- N.A

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**(E) cGMP REQUIREMENTS****1.0 Material of construction specifications :**

- Any product contact parts shall be SS 316/SS 316 L and Non contact surface shall be of SS 304.
- Equipment shall be designed to facilitate easy and thorough Cleaning
- Seals, gaskets and lubricants if used shall be of Food Grade.

**2.0 Working area environment requirement :**

- Area: As per ISO Class "8"/100000.
- Temperature: NMT 27°C (15°C to 27°C)
- RH: NMT 60% RH

**3.0 Others :**

- MOC / polish of pan shall be such that tablets when coated with suspension containing Titanium Dioxide and Talc will not exhibit black spots due to rubbing or friction.
- Process air must be filtered through **HEPA** filter. 5 MICRON FILTER PROVIDED WITH MACHINE. HEPA NOT REQUIRED
- All contact parts must be of smooth finish & easy to clean. All joints must be smoothly finished.

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**(F) DOCUMENTATION REQUIREMENTS:**

User manual,  
 DQ,IQ, OQ PROTOCOL  
 warranty certificate  
 GA drawing, PID, Electrical drawing-as applicable  
 MOC certificate  
 Calibration certificate  
 Food grade certificate

**(G) SAFETY REQUIREMENTS:**
**1) Safety related to operator:**

- Electric panel connection shall be properly sealed and be flameproof. NON FLAMEPROOF PANEL REQUIRED
- Emergency stop button. ALL MOTORS TO BE FLAMEPROOF
- Noise level shall not exceed than 80 db
- Earthing point facility.

**2) Safety related to process**

- N.A

**3) Process control Alarms/ Interlocks:**

- List of alarm / interlock shall be provided by vendor along with suitable message.
- Alarm system for lack of utilities during run period of auto coater.
- Air Pressure Minimum
- Inlet Temperature < minimum
- Inlet Temperature > maximum
- Exhaust temperature < minimum
- Exhaust temperature > maximum
- Inlet Blower drive error

## USER REQUIREMENT SPECIFICATION [RESTRICTED CIRCULATION]

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- Product Bed Temperature < minimum
- Product Bed Temperature > maximum
- Process Time End
- DP pan minimum / maximum
- Atomisation air pressure < minimum
- Inlet air CFM minimum
- Pan motor drive error
- DP HEPA inlet > maximum
- DP HEPA inlet < minimum

• Interlock messages :

- Pan Motor tripped
- Inlet blower tripped
- Exhaust motor tripped
- Peristaltic pump motor tripped NOT POSSIBLE
- Machine guard open SIDE DOOR INTERLOCK TO BE PROVIDED
- Emergency stop pressed
- AHU door open

4) Password protection:

- Password protections are required with appropriate security (compatible 21 CFR part 11 complaints) at different (Three) levels like- 21CFR NOT REQUIRED. NO PRINTER AND DATA LOGGING REQUIRED
  - Operator
  - Supervisor
  - Manager
- **Operator:** Operational access only, including start, stops, reset & default correction. No data transfer is allowed.
- **Supervisor:** All access of operator, including approvals transfer of data to hierarchical systems.

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- **Manager:** Data accessibility of both operator and supervisor, including recipe systems. Change the password of operator, supervisor levels and himself level.
- Individual log-in password system shall be required, i.e. 3 individual log-ins in level-I & II and 2 individual log-ins in level-III.

#### (H) ABBREVIATIONS:

| Abbreviation       | Full Form                            |
|--------------------|--------------------------------------|
| AHU                | Air Handling Unit                    |
| CFM                | Cubic Feet Per Meter                 |
| CFR                | Code of Federal Regulation           |
| cGMP               | Current Good Manufacturing Practices |
| °C                 | Degree Centigrade                    |
| CIP                | Cleaning In Place                    |
| db                 | Decibel                              |
| DQ                 | Design Qualification                 |
| GA                 | General Arrangement                  |
| HEPA               | High Efficiency Particulate Air      |
| ID No.             | Identification Number                |
| IQ                 | Installation Qualification           |
| kg.                | Kilogram                             |
| kg/cm <sup>2</sup> | Kilogram Per Centimetre Square       |
| mm                 | Millimetre                           |
| MOC                | Material of Construction             |
| NMT                | Not More Than                        |
| No.                | Number                               |
| OQ                 | Operational Qualification            |
| P&I                | Piping and Instrumentation           |
| RH                 | Relative Humidity                    |
| RPM                | Revolution Per Minute                |
| SS                 | Stainless Steel                      |
| URS                | User Requirement Specification       |
| FAT                | Factory Acceptance Test              |

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**(I) DISCUSSION / REVIEW / COMMENTS:**

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## USER REQUIREMENT SPECIFICATION [RESTRICTED CIRCULATION]

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**(J) USER REQUIREMENT SPECIFICATION APPROVAL:**

|             | NAME             | DESIGNATION         | SIGN       | DATE       |
|-------------|------------------|---------------------|------------|------------|
| Prepared By | Poochiksha Nayak | ATM                 | Poochiksha | 07/04/2015 |
| Reviewed By | Aruny Patel      | Am                  | Aruny      | 05/04/2015 |
|             | Arpita Patel     | Manager<br>AB Patel | Arpita     | 07/04/2015 |
|             | Rahul Singam     | Manager             | Rahul      | 07/04/2015 |
|             | Rahul Dixit      | Asst                | Rahul      | 07/04/2015 |
|             | Yogesh Patel     | AGM                 | Yogesh     | 07/04/2015 |
|             | J.G. Thakur      | AGM                 | J.G.       | 07/04/2015 |
| Approved By | Krishna Shavanna | VP                  | Krishna    | 07/04/2015 |